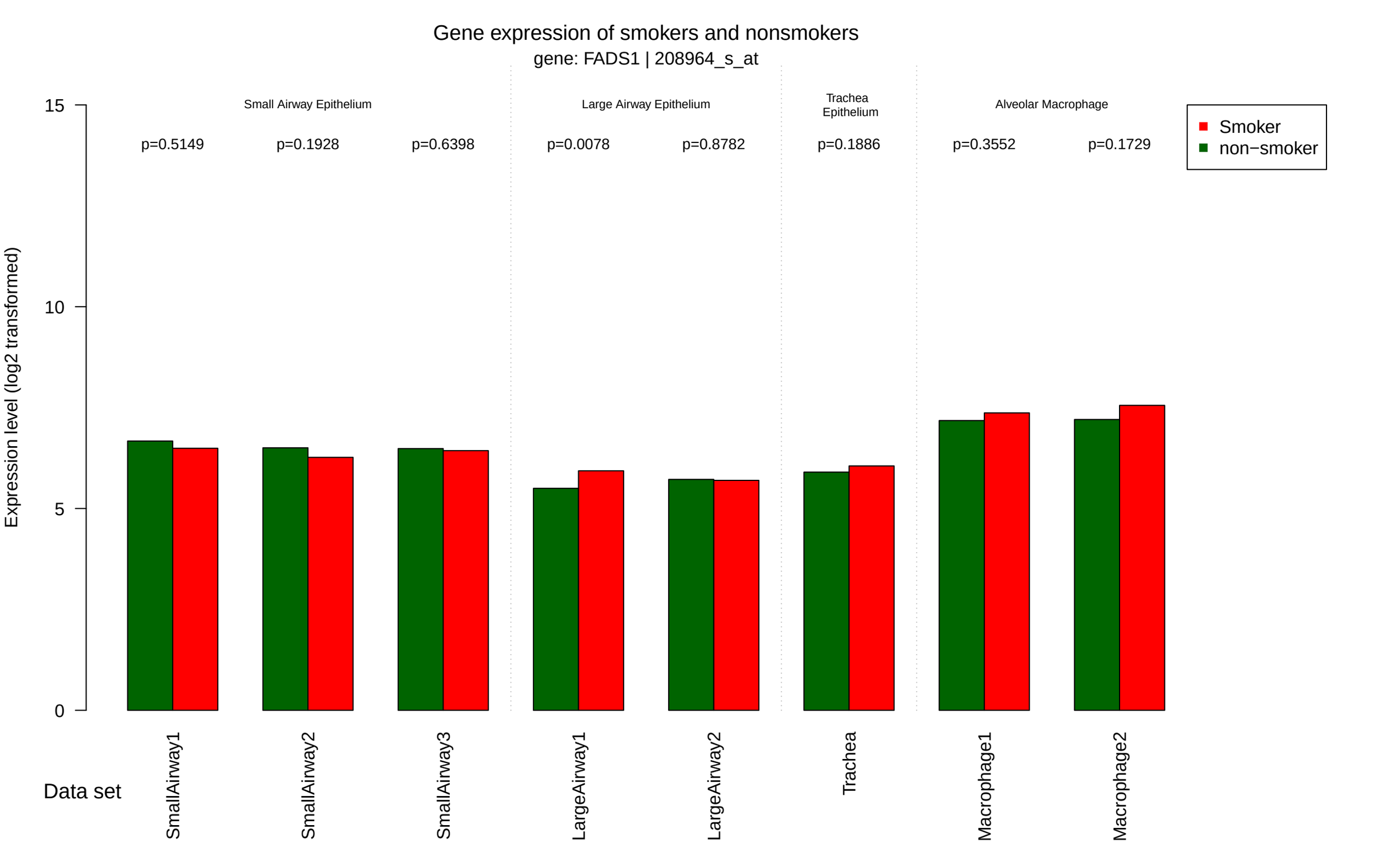
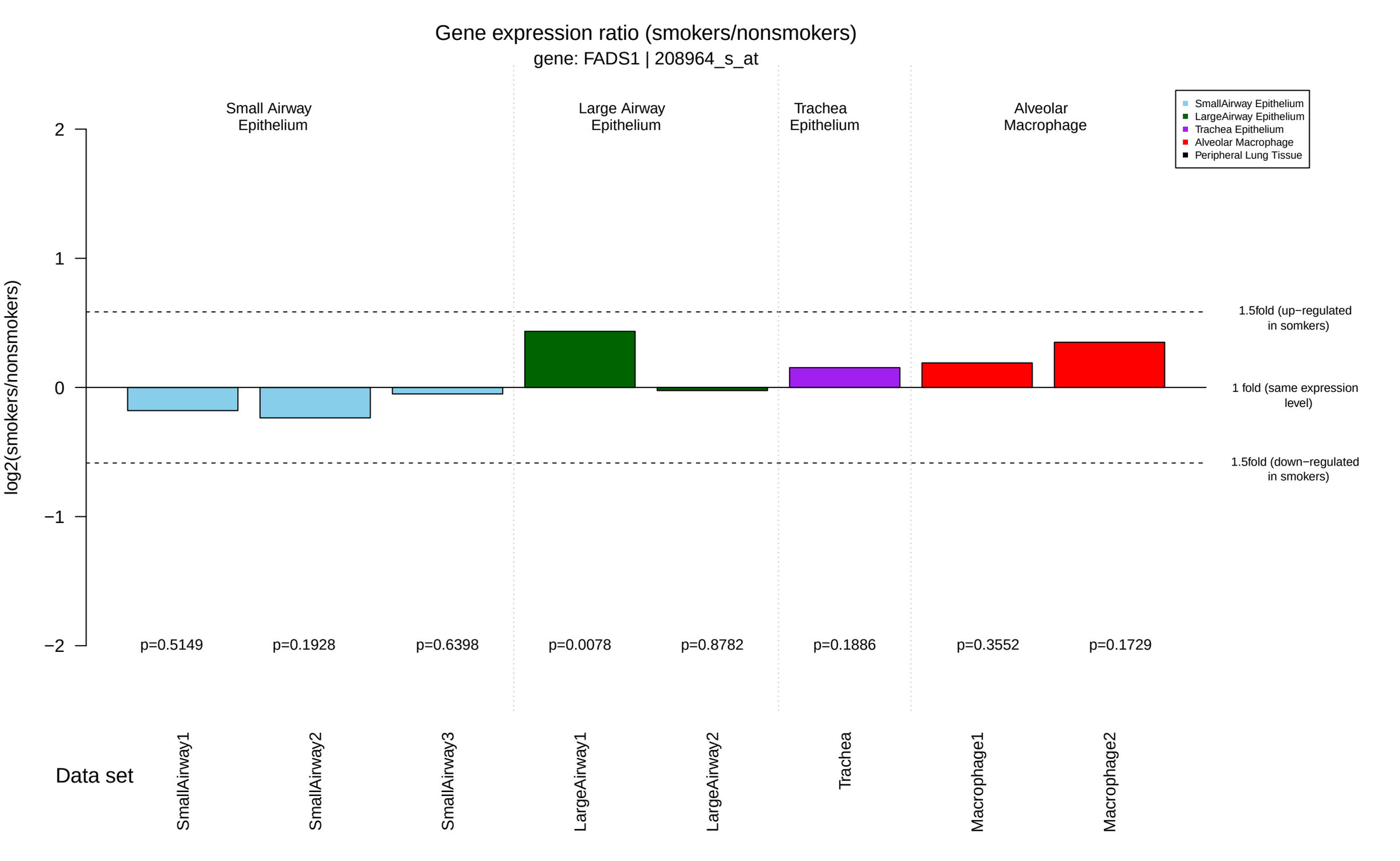
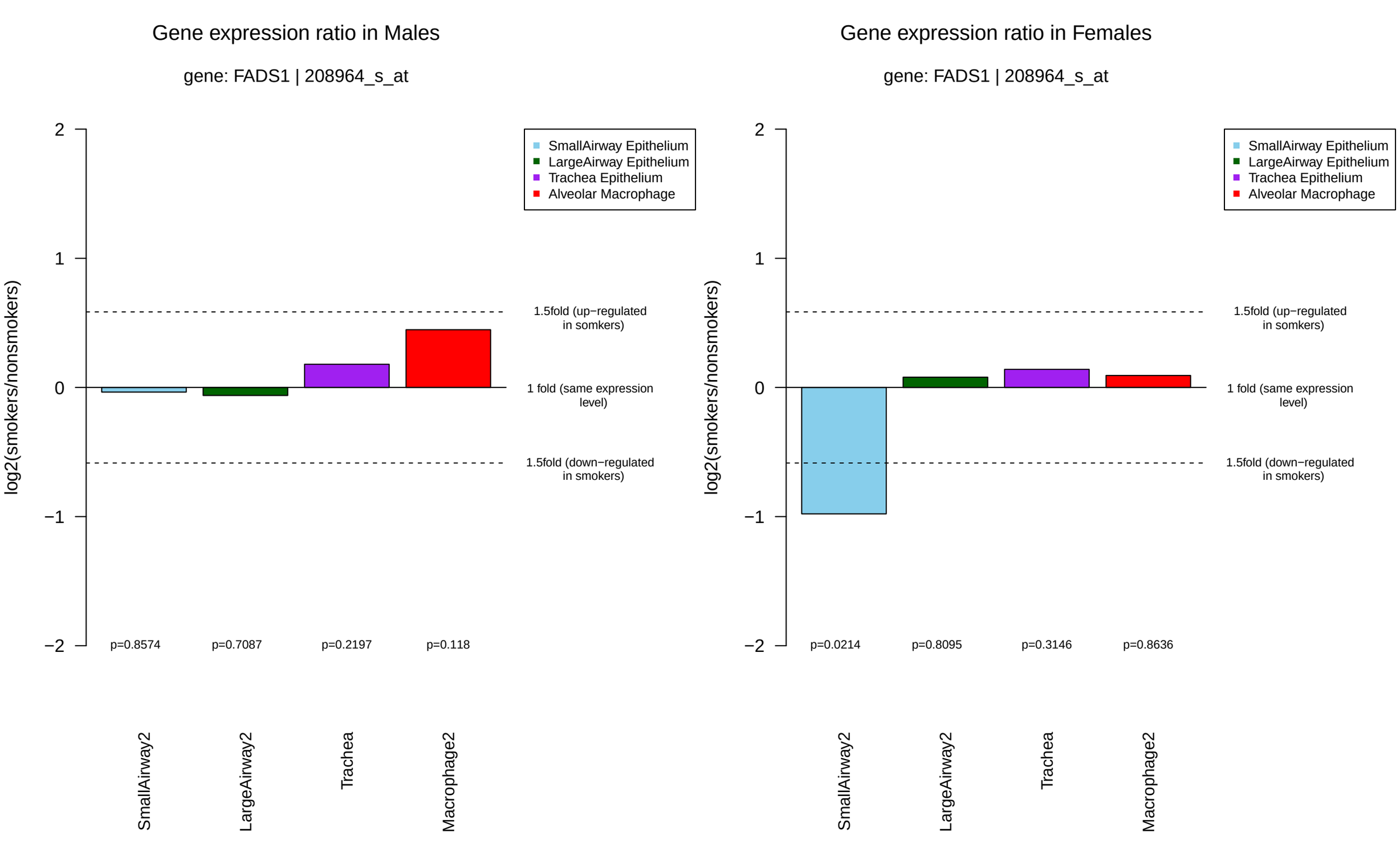
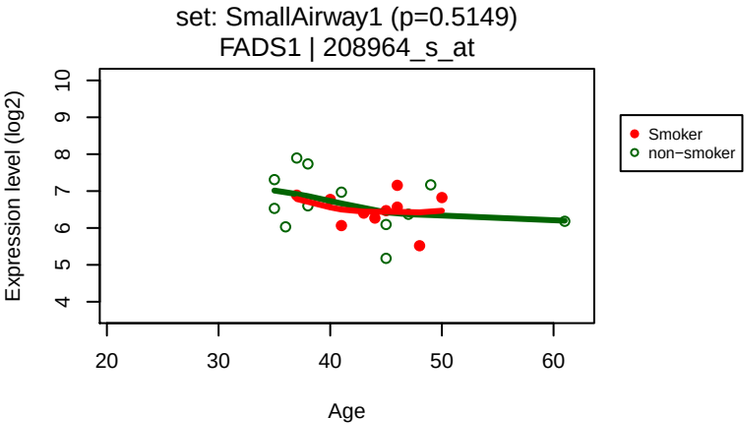
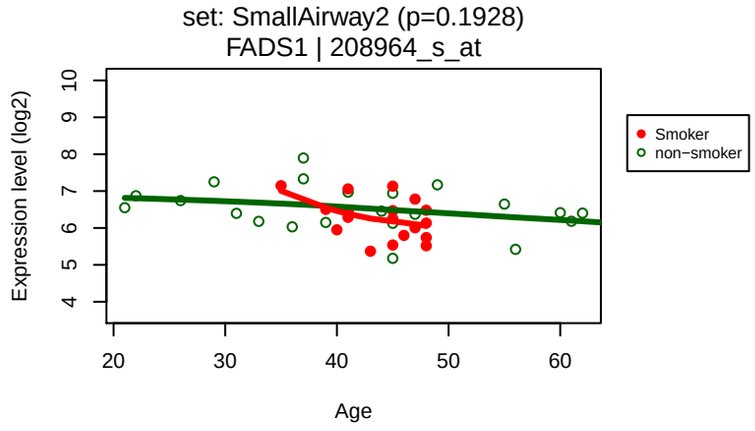
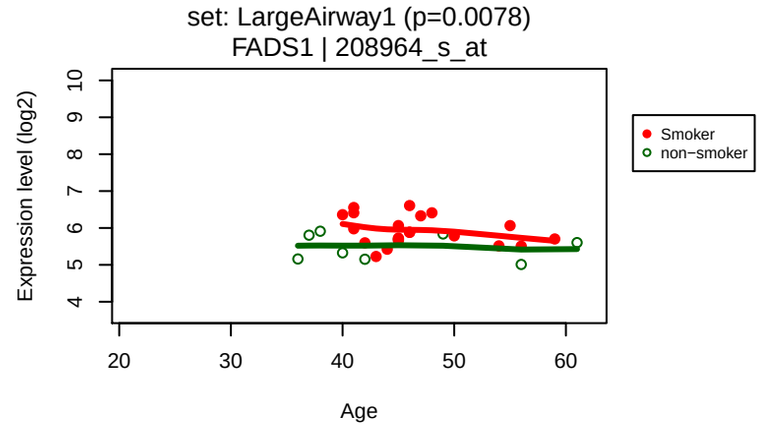

Gene expression of smokers and nonsmokers
gene: FADS1 | 208964_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5149
p=0.1928
p=0.6398
p=0.0078
p=0.8782
p=0.1886
p=0.3552
p=0.1729
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

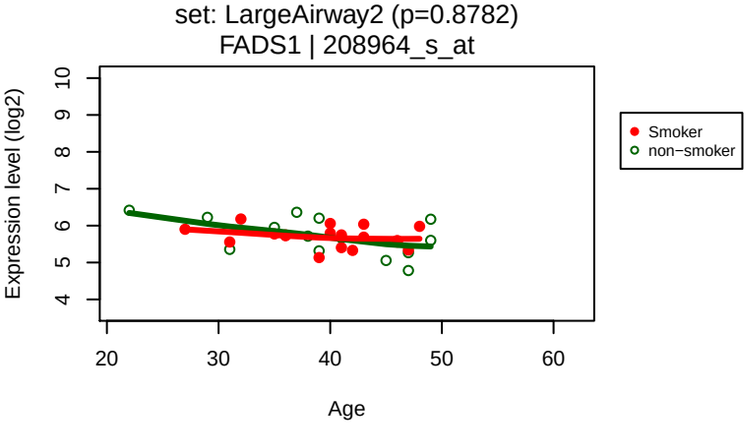

Gene expression ratio (smokers/nonsmokers)
gene: FADS1 | 208964_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5149
p=0.1928
p=0.6398
p=0.0078
p=0.8782
p=0.1886
p=0.3552
p=0.1729
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

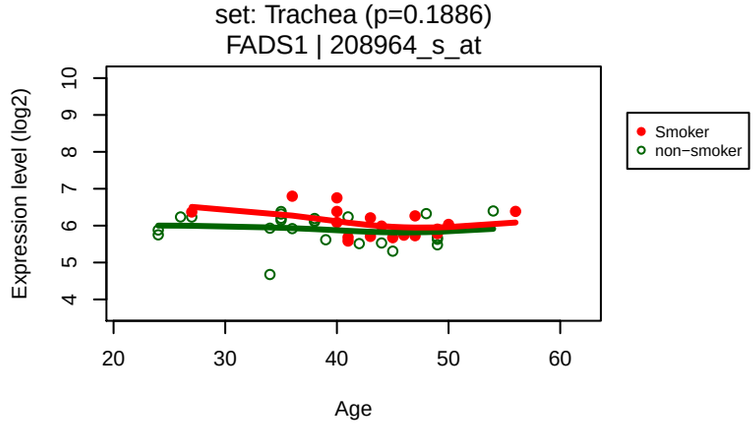

Gene expression ratio in Males
gene: FADS1 | 208964_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8574
p=0.7087
p=0.2197
p=0.118
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FADS1 | 208964_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0214
p=0.8095
p=0.3146
p=0.8636
SmallAirway2
LargeAirway2
Trachea
Macrophage2

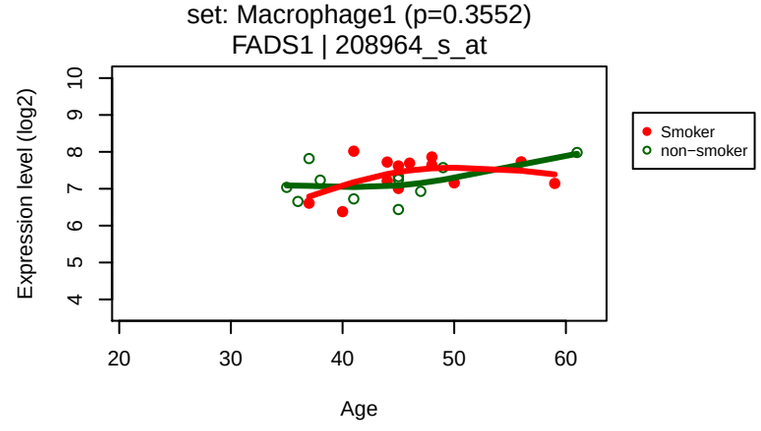

set: SmallAirway1 (p=0.5149)
FADS1 | 208964_s_at
Expression level (log2)
Age
Smoker
non−smoker

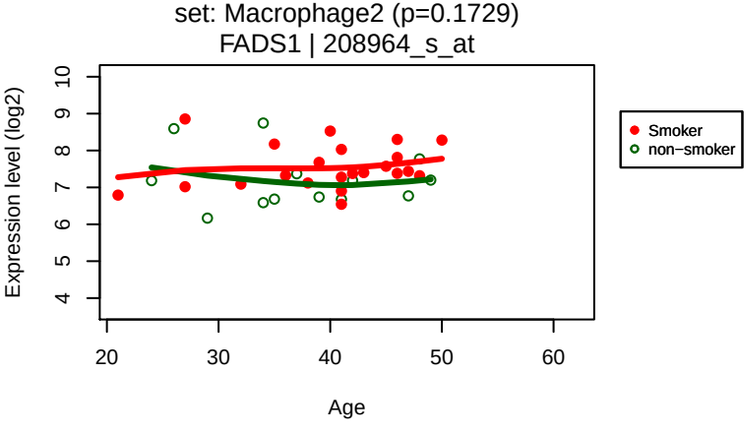

set: SmallAirway2 (p=0.1928)
FADS1 | 208964_s_at
Expression level (log2)
Age
Smoker
non−smoker

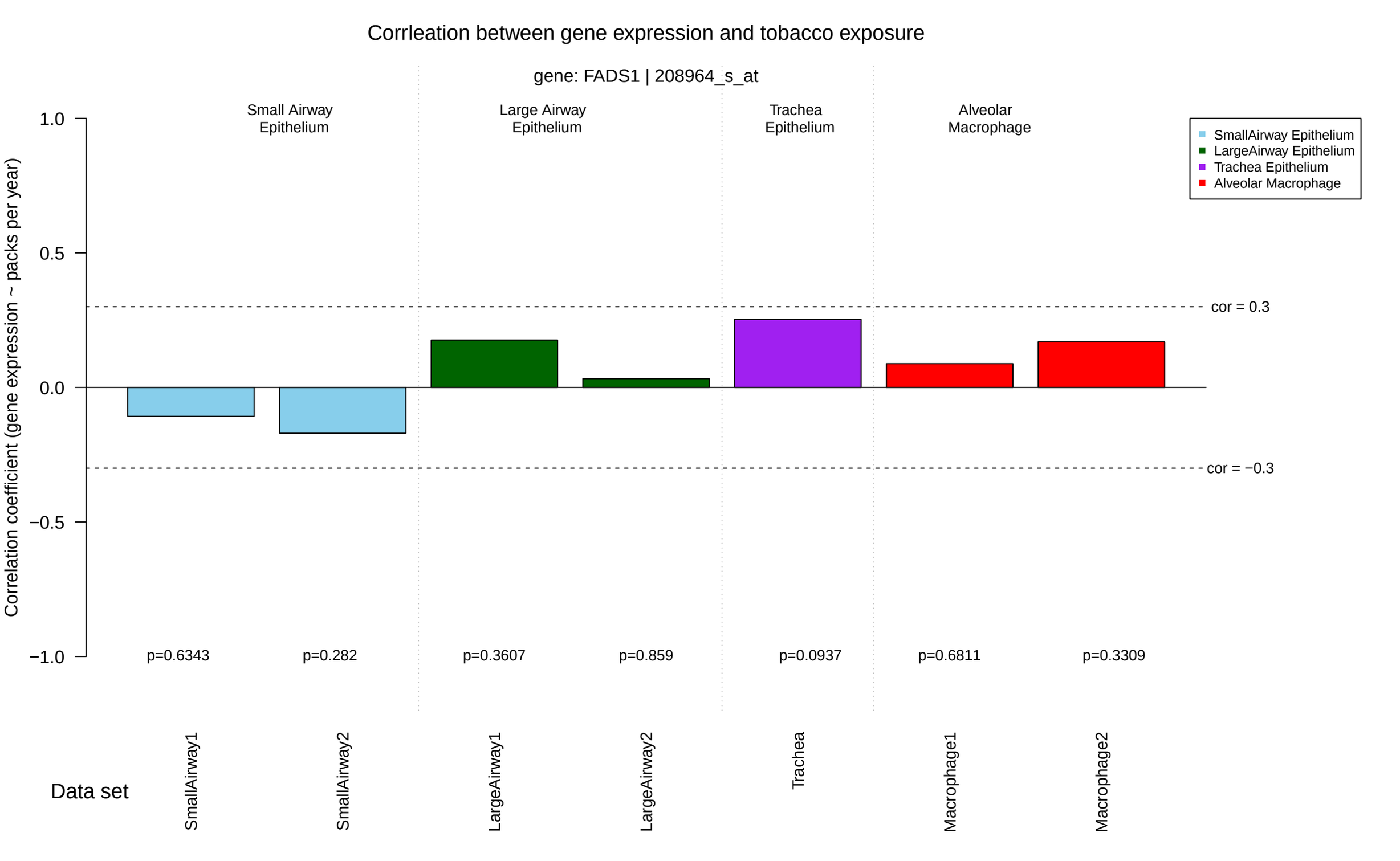

set: LargeAirway1 (p=0.0078)
FADS1 | 208964_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8782)
FADS1 | 208964_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1886)
FADS1 | 208964_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3552)
FADS1 | 208964_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1729)
FADS1 | 208964_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FADS1 | 208964_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6343
p=0.282
p=0.3607
p=0.859
p=0.0937
p=0.6811
p=0.3309
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2